Renew — Full Review

FORMULA

MOTS-c 10mg + SS-31 10mg + NAD+ 500mg

What's in it

PEPTIDE	DOSE	ROLE IN THIS BLEND
MOTS-c	10mg	The signal piece. Activates AMPK, the cell's main metabolic switch, telling mitochondria to multiply and burn fuel more efficiently.
SS-31	10mg	The structural piece. Binds and stabilizes cardiolipin in the inner mitochondrial membrane — the only ingredient here with a real FDA approval behind it (as elamipretide, for Barth syndrome).
NAD+	500mg	The fuel piece. The coenzyme every one of those energy-producing reactions actually runs on — levels decline substantially with age.

The combination logic

Each of these three works on a different part of the same problem. SS-31 repairs the physical structure inside the mitochondria where energy is made. MOTS-c sends a signal that tells the cell to build more mitochondria and use fuel more efficiently. NAD+ supplies the raw material every one of those energy-producing reactions actually runs on. Put together, the idea is structure, signal, and fuel, addressed at the same time.

Clinical detail: SS-31 (elamipretide) selectively binds cardiolipin in the inner mitochondrial membrane, protecting cristae architecture and electron transport chain function. MOTS-c is a mitochondrial-DNA-encoded peptide that activates AMPK signaling, promoting mitochondrial biogenesis and improved glucose/fatty-acid metabolism. NAD+ is the essential coenzyme substrate for Complex I of the electron transport chain and for sirtuin activity; its cellular pool declines significantly with age. The three are proposed to act on non-competing points of the same energy-production pathway.

Is there evidence for the combination, or only for the parts?

> SS-31 has the most developed evidence of the three, including FDA accelerated approval (as elamipretide) for a rare mitochondrial condition, Barth syndrome — though that approval does not extend to general energy or longevity use. MOTS-c has meaningful animal data on metabolic adaptation and preliminary human interest. NAD+ has a large body of general metabolic literature but is a coenzyme, not a drug, and dosing conventions vary widely by route. No study has tested this specific three-part combination.

What it's used for

USED FOR	EVIDENCE	IN SHORT
Cellular energy / fatigue support	animal_only / limited human	The most consistent rationale across the three components — each targets a distinct point in mitochondrial energy production.

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Metabolic and age-related decline	animal_only	MOTS-c in particular has been studied for metabolic adaptation and insulin sensitivity in animal models.
Mitochondrial membrane protection	human_limited	SS-31 (elamipretide) has the most developed human data of the three, including an FDA-approved use for a rare mitochondrial disease.

Overlaps & stacked risks

- Each component carries its own risk profile; combining them does not reduce that.
- MOTS-c and NAD+ both interact with glucose/insulin metabolism — monitor in patients with diabetes.
- Rapid NAD+ administration is the most common source of acute discomfort (flushing, chest tightness) in this combination.
- Long-term human safety data for this specific combination does not exist.

Who it’s not for

- Pregnant or breastfeeding patients
- Patients with significant cardiac or renal disease, without clinician oversight
- Anyone on blood-sugar-lowering medication, due to MOTS-c’s effect on glucose metabolism
- Patients expecting a fast-infusion NAD+ experience — rapid administration can cause transient flushing or chest tightness

Who it suits

Patients pursuing general energy, metabolic, or age-related support who have discussed the individual evidence profile of each component (strongest for SS-31, more preliminary for MOTS-c and NAD+ dosing) with their provider.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

This page describes Renew for general educational purposes. It is not medical advice and does not replace a consultation with a licensed provider. Individual results vary, and combination products carry the risks of each included peptide.